DEEPFAKE1

Detection and Forensic Analysis Using Machine Learning

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Team Contributions

The following table summarizes each team member's algorithm, training dataset, and primary contribution to this project.

Member	Algorithm	Training Dataset	Contribution
Aena	EfficientNet-B4	FaceForensics++ (FF++) C23	Training, cross-dataset testing,
Sarra	MesoNet	UADFV	Training, cross-dataset testing
Lilas	XceptionNet	CelebDF_V2(image Dataset)	Training, cross-dataset testing
Bhumi	ResNet-50	CIFAKE	Training, cross-dataset testing
Meriem	ViT	Open Forensics	Training, cross-dataset testing

Abstract

This report presents a comparative evaluation of five deepfake detection algorithms implemented and tested. The core objective is to identify which algorithm is most reliable and suitable for real-world cybercrime investigation, particularly in scenarios where a forensic investigator encounters deepfake evidence from an unknown source. Results are analyzed to determine which algorithm generalizes best to unseen data, as this is the most critical property for a forensic tool used in criminal investigations.

1. Introduction

Deepfakes are synthetic media generated using deep learning techniques such as autoencoders and Generative Adversarial Networks (GANs) to manipulate or replace facial identities in images and videos. Over the past few years, deepfakes have become increasingly accessible and convincing, making them a serious tool for fraud, identity theft, political manipulation, and non-consensual content. For cybercrime investigators, the ability to reliably detect deepfake content is no longer optional but essential.

The central challenge in deepfake forensics is not simply building a model that performs well in a controlled setting, but finding an algorithm that generalizes to unseen, real-world content. A forensic investigator cannot choose what kind of deepfake a suspect used or what dataset it resembles. The detection tool must work regardless of the source.

This project addresses that challenge directly. Five algorithms were each trained on a different deepfake dataset and then tested on all other team datasets, simulating a real forensic scenario where the model encounters unknown content. The goal is to determine which algorithm generalizes best across datasets, making it the most trustworthy candidate for cybercrime investigation. Findings are then applied to a simulated client case and analyzed within the Canadian legal framework for presenting digital evidence in court.

2. Methodology

2.1 Datasets

Each team member selected a publicly available deepfake benchmark dataset for training. The following datasets were used:

- **FaceForensics++** at C23 compression is one of the most widely adopted benchmark datasets in deepfake detection research. It contains video-based face manipulations generated using four distinct methods: DeepFakes, Face2Face, FaceSwap, and NeuralTextures, providing a diverse range of manipulation techniques within a single dataset. The dataset was obtained via Kaggle (xdxd003/ff-c23) using the Kaggle CLI and stored on the Concordia Speed HPC cluster.

  The C23 compression level closely mirrors real-world video sharing platforms where deepfake content is most commonly encountered, making it highly relevant for forensic applications. Its widespread use in published literature also allows our results to be directly compared against established benchmarks, strengthening the scientific validity of our findings
- **UADFV** (University at Albany DeepFake Video dataset) is a benchmark dataset for deepfake detection, designed to study how well models can distinguish between real and manipulated (fake) videos of human faces.
- **CelebDF_V2**(image Dataset) [16] is a high-quality deepfake benchmark derived from the original Celeb-DF (V2) videos. Faces were detected from the original dataset Celeb-DF (V2) using dlib's frontal face detector. Faces were cropped out and then enhanced using ESPCN-based super-resolution, yielding 50,740 real and 50,751 fake images, split into three categories for training, validation and testing. All images were resized to 256 x 256 pixels. Celeb-DF (V2) dataset is challenging for deepfake detection since the deepfakes were refined by fixing common visual artifacts such as low resolution, color mismatch, and temporal flickering [13].
- **CIFAKE** - CIFAKE is a large-scale dataset containing 120,000 images (60,000 real and 60,000 synthetic).Real Images: Sourced from the CIFAR-10 dataset(natural objects, animals, vehicles).Fake Images: Generated using Stable Diffusion (v1.4) to match the categories of CIFAR-10.Format: All images are 32*32 pixels, which allowed for rapid batch processing.
- **OpenForensics**- OpenForensics [5] is a large-scale challenging dataset for face forgery detection. It is designed to reflect real-world (in the wild) conditions, including variations in lighting, pose, occlusion, and image quality [5]. It contains both authentic and manipulated facial media, covering different types of deepfake techniques such as face swapping and facial reenactment [5]. In this work, we use the Kaggle version, which is a processed, pre-extracted and labeled images, organized into two classes: real and fake.The dataset is split into 140k+ images for training and 10,905 images for testing.

All datasets were stored in the shared team directory on the Concordia Speed HPC cluster at: /speed-scratch/inse6610team2/

2.2 Algorithms

Each team member implemented a distinct deep learning architecture for binary classification (real vs. fake):

- **EfficientNet-B4** is a highly efficient convolutional neural network that achieves strong performance by uniformly scaling depth, width, and resolution using a compound scaling method. It was pre-trained on ImageNet, providing robust general-purpose feature extraction capabilities that transfer well to deepfake detection tasks. For this project, EfficientNet-B4 was selected due to its superior accuracy-to-parameter ratio, making it well suited for detecting subtle manipulation artifacts in compressed face images.
- **MesoNet** is a lightweight CNN designed for deepfake detection. It focuses on mesoscopic features, which capture intermediate-level image artifacts introduced during face manipulation. .
- **XceptionNet** [17] is a convolutional neural network (CNN) inspired by Inception, where it replaces standard convolutions with depthwise separable convolutions. The depthwise separable convolution is a process divided into two parts. It applies a separate filter to each channel of the image which is the depthwise convolution, then it is followed by the pointwise convolution where it uses a 1x1 filter to put the results from all channels together. This allows the network to detect subtle inconsistencies that may be invisible to the human eye. The model used in this project was initialized with pre-trained ImageNet weights and then fine-tuned for binary classification (real/fake).
- **ResNet-50** was initially considered, but ResNet-18 was selected for its superior environment setup speed for testing. It offers a high accuracy-to-parameter ratio, making it ideal for the 32*32 resolution of CIFAKE, where a 50-layer network might lead to overfitting.It utilizes 8 residual blocks. Each block employs a shortcut connection that skips one or more layers, effectively solving the vanishing gradient problem.
- **Vision Transformer** (ViT) [6] - ViT is a transformer-based architecture originally introduced for image classification, which applies the self-attention mechanism [6]. Unlike convolutional neural networks (CNNs), ViT processes an image by dividing it into a sequence of fixed-size patches. Each patch is flattened and linearly embedded into a vector representation. Positional embeddings are added to retain spatial information, and the resulting sequence is fed into a transformer encoder composed of multi-head self-attention layers. We selected ViT as it is able to capture global relationships across the entire image using self-attention [9]. Deepfake manipulations often introduce subtle and spatially distributed artifacts that are not confined to local regions [1]. Unlike convolutional models that focus on localized patterns, ViT analyzes the image holistically, enabling it to better detect these inconsistencies [8]. This makes it particularly effective for identifying sophisticated deepfake manipulations in real-world scenarios.

  In this work, we use a pretrained ViT model from the Hugging Face. You can find all the original ViT checkpoints under the Google organization here.

2.3 Environment Setup - Concordia Speed HPC Cluster

All training and testing was conducted on the Concordia University Speed High Performance Computing (HPC) cluster. The key environment details are:

- Cluster: speed-submit.encs.concordia.ca
- GPU: NVIDIA Tesla V100
- Job Scheduler: SLURM
- Python Environment: Anaconda 2023.03, PyTorch 2.2.2+cu118
- Shared Team Directory: /speed-scratch/inse6610team2/

Jobs were submitted using SLURM batch scripts (.sh files) and monitored via squeue. The conda environment was activated using the full path:

2.4 Individual Training Methodology and Results

EfficientNet-B4 on FaceForensics++ (FF++) C23

EfficientNet-B4 was trained for 10 epochs on face images extracted from the FF++ dataset at C23 compression. A total of 6,000 images were used, split into 70% training, 15% validation, and 15% test sets. Training was conducted on the Concordia Speed HPC cluster using JupyterHub with a Tesla V100 GPU, with jobs submitted and monitored via SLURM batch scripts.

Partial layer freezing was applied to the first three convolutional blocks to preserve low-level features learned during ImageNet pre-training, while allowing the deeper layers to adapt specifically to deepfake detection patterns present in FF++ C23. Binary Cross-Entropy with Logits was chosen as the loss function and the Adam optimizer was used with a learning rate of 1e-4 and weight decay of 1e-6 to reduce overfitting. The model converged steadily over 10 epochs with the best model checkpoint saved based on validation accuracy. Full training results are available in the comparison table in Section 3.

- Optimizer: Adam (lr=1e-4, weight_decay=1e-6)
- Loss Function: Binary Cross-Entropy with Logits
- Image Size: 224x224
- Batch Size: 64
- Epochs: 10

MesoNet on UADFV

MesoNet (Meso4) was trained for 10 epochs on image frames extracted from the UADFV dataset. The dataset was split into 70% training, 15% validation, and 15% test sets. Approximately 660 training images were used (about 334 real and 327 fake). Unlike transfer learning models, MesoNet was trained from scratch without layer freezing, focusing on learning mesoscopic-level artifacts associated with deepfake manipulations.

- Training Accuracy: 95.16%
- Val Loss: 14,66 | Val Acc: 92.96
- Val AUC: 98.41%
- Optimizer: Adam (lr=1e-4)
- Loss Function: Binary Cross-Entropy with Logits
- Image Size: 256 × 256
- Batch Size: 8

XceptionNet on CelebDF_V2(image Dataset)

Dataset and Preprocessing

The XceptionNet model was trained and evaluated on CelebDF_V2(image Dataset) [16] which is an image version of the original Celeb-DF (V2) video dataset [13]. The dataset is split into three folders for training (Train), validation (Val) and testing (Test) where each of them is divided into two subfolders named "fake" and "real" respectively:

- Train: 80,824 images (fake: 40536, real: 40288),
- Val: 10,104 images (fake: 5068, real: 5036),
- Test: 10,103 images (fake: 5067, real: 5036).

All images were resized to 299 x 299 pixels since this is the input size expected by XceptionNet. The training images were automatically labeled by Keras based on folder names. The "fake" folder was assigned class 0 and the "real" folder was assigned class 1 by default since the folder names are sorted alphabetically.

Data augmentation

Small random changes were applied on the training images to reduce the possibility of the model memorizing the training images and to help the model learn better. This is the process known as data augmentation. The images used for validation and testing did not go through that process, thus stayed unchanged. The changes were:

- Random rotation up to ±20 degrees,
- Random horizontal and vertical shifting up to 20% of the image size,
- Random horizontal flipping, producing mirror images.

These small variations make the model more robust and similar to real-world differences such as angle, orientation, etc.

Model Architecture and Training Setup

The model used the XceptionNet architecture which was pre-trained on ImageNet with the top classification layer removed which is considered the final layer that outputs the predicted class. A new head has been added on top instead to make the model do binary classification instead (real/fake) which consists of:

- Global Average Pooling to reduce the 2D feature maps to a single vector (1D),
- One dense layer with 128 neurons and ReLU activation to learn deepfake-specific patterns,
- Final output layer with 1 neuron and sigmoid activation to output probabilities between 0 and 1.

The base XceptionNet layers were frozen at the start so the model could learn from the ImageNet features before adding our own layers for fine-tuning.

Training was performed on the Concordia Speed HPC cluster using an NVIDIA A100-SXM4-80GB MIG 2g.20gb GPU with these specific configurations:

- Framework: TensorFlow/Keras (version 2.21.0),
- Precision: Mixed precision (mixed_float16) to save GPU memory,
- Batch size: 16 images at a time,
- Epochs: 10,
- Optimizer: Adam (default)
- Loss function: Binary cross-entropy
- Metrics tracked during training: Accuracy, AUC-ROC and Loss.

Training

During the training over 10 epochs, the model's accuracy steadily improved while the validation loss decreased which shows the model was learning properly without major overfitting.

Below are the metrics at the first, fifth and last epoch (the full per-epoch logs are available in the "xception_deepfake.ipynb" file), showing the model's growth.

Epoch	**Train Accuracy**	**Train AUC-ROC**	**Train Loss**	**Validation Accuracy**	**Validation AUC-ROC**	**Validation Loss**
1	0.6500	0.7113	0.6145	0.6836	0.7705	0.5789
5	0.7385	0.8170	0.5132	0.7370	0.8264	0.5194
10	0.7592	0.8411	0.4826	0.7579	0.8495	0.4851

Training accuracy increased from 65.00% to 75.92&. Validation accuracy rose from 68.36% to 75.79%. Both the training and validation loss decreased consistently and the validation curves closely followed the training curves.

All relevant files related to this experiment are available in the repository under "src/team2/xceptionnet_celebdf".

ResNet on CIFAKE

- Dataset and Preprocessing:

For my contribution, I utilized the CIFAKE dataset, comprising 120,000 images divided into real and synthetic classes. The preprocessing pipeline was implemented using the Torchvision transforms library to standardize the input for the ResNet architecture.

- Image Transformation:

Although the native resolution of CIFAKE is 32 x 32, the images were resized to 224 x 224 pixels to remain compatible with the ResNet-18 input requirements.

- Normalization:

Standard ImageNet normalization parameters (mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]) were applied to the tensors to ensure the input data distribution matched the pre-trained model expectations.

- Optimization Strategy:

I selected ResNet-18 using the IMAGENET1K_V1 weights. This approach was chosen to maximize environment setup speed on the Concordia Speed HPC cluster, as the 18-layer architecture provides a faster training turnaround compared to ResNet-50 while still offering high feature extraction capabilities via transfer learning.

Hyperparameters:

Optimizer: Adam (learning rate = 0.0001)

Loss Function: Cross-Entropy Loss
Batch Size: 32
Epochs: 3

All relevant files can be located under: "/src/team2/CIFAKE-ResNet"

ViT on OpenForensics

Architecture: We use ViT (ViT-Base-16) architecture to train our deepfake image detection model on the OpendForensics dataset. We followed the data directory split by using the train folder for model training and the test folder for the evaluation. In our setup, the training set contains 140k+ images classified into two classes including real and fake (70k image per class). The problem is a binary classification problem.
ViT processes an input image by dividing it into fixed-size 16×16 patches, converting these patches into a sequence of embeddings, and then modeling their relationships through Transformer layers based on self-attention. In particular, the ViT-B16 configuration expects images resized to 224×224 pixels and uses 16×16 patch resolution during processing. This allows the model to capture subtle semantic artifacts in fabricated imagery that local filters often miss.
Training Metrics and Hyperparameters: Due to the large-scale dataset and the efficiency of the Transformer architecture, the model converged within 3 epochs. A total of 11,814 global steps were completed, processing a diverse set of facial samples. Early convergence indicates effective transfer learning, while additional epochs may increase the risk of overfitting.

Job Script: /speed-scratch/f_meriem/test.sh

Test Script: /speed-scratch/f_meriem/test_model.py

Optimizer: Adam (lr = 2e−5, ε = 1e−8)
Loss Function: Binary Cross-Entropy with Logits
Image Size: 224 × 224
Batch Size: 32
Epochs: 3

2.5 Testing Methodology

EfficientNet-B4 on FaceForensics++ (FF++) C23

The testing script (test.py) was executed as a SLURM batch job on the Concordia Speed HPC cluster using a Tesla V100 GPU. The same preprocessing pipeline used during training was applied during testing:

- Image resized to 224x224 pixels
- Normalized with ImageNet mean [0.485, 0.456, 0.406] and std [0.229, 0.224, 0.225]
- No data augmentation applied during testing
- Classification threshold: 0.5 (sigmoid output >= 0.5 classified as fake)
- Batch size: 64, evaluated using GPU acceleration
- Metrics computed: Accuracy and AUC-ROC per dataset

Job Script: /speed-scratch/ae_verma/test_job.sh

Test Script: /speed-scratch/ae_verma/test.py

MesoNet on UADFV

The trained MesoNet model was evaluated on the UADFV test set, which consists of image frames that were not used during training. This ensures a fair assessment of the model's performance on unseen data from the same dataset .

The testing script test_model.py was executed as a SLURM batch job on the Concordia Speed HPC cluster same preprocessing pipeline used during training was applied during testing

Job Script: /speed-scratch/s_benred/test.sh

Test Script: /speed-scratch/s_benred/test_model.py

XceptionNet (Cross-Dataset Evaluation)

The XceptionNet model trained on CelebDF_V2(image Dataset) [16] was used to evaluate on five different datasets without additional training or fine-tuning. This process is to assess the model's ability to detect deepfake images beyond the dataset that was used for its training which is very important for real-world cybercrime investigations.

<u>Purpose</u>

A model that only performs well on its training dataset is not reliable for forensic use since investigators do not know necessarily how the deepfake images were generated, what tools were used and what methods were implemented to create deepfakes. Cross-dataset testing can evaluate the true robustness of the model.

<u>Testing Procedure</u>

For every dataset the same saved model was used. The testing procedure was the same for all datasets:

- Images were loaded using ImageDataGenerator with rescaling,
- All images were resized to 299 x 299 pixels.
- The model produced a probability for each image.

- A threshold of 0.5 was applied where a probability higher than 0.5 means a predicted real image (class 1) while a probability equal to or below 0.5 means a predicted fake image (class 0),
- The evaluation metrics were Accuracy, AUC-ROC, F1-Score and Log Loss which were calculated,
- A confusion matrix was generated for each tested dataset.

All relevant files of these experiments can be found in the repository under "src/team2/xceptionnet_celebdf".

ResNet on CIFAKE

Testing Procedure: The testing phase utilized a separate inference script executed as a SLURM batch job to verify the model's performance on both native and external datasets.

Preprocessing: The testing script mirrored the training transformations, including the 224 x 224 resizing and ImageNet normalization, to maintain feature consistency across all evaluations.

Inference and Classification: The model was set to evaluation mode to disable dropout and batch normalization updates. The classification was determined by selecting the highest probability from the two-feature output of the final linear layer.

All relevant files can be located under: "/src/team2/CIFAKE-ResNet"

ViT on Open Forensics

First, we evaluate the trained model on the test data from OpenForensics data, which contains 10k+ (5492 fake and 5413 real) images.

Then, we implement a multi-level benchmark:

1. Cross-Dataset Evaluation: we benchmarked our Vision Transformer against 6 distinct test sets (including Celeb-DF, CIFAKE, and FaceForensics++), totaling ~87,000 images.
2. Explainability Analysis: We extracted self-attention weights from the final Transformer layer to provide forensic evidence.

3. Global Benchmark Test: The model was specifically validated against
the 140k Global Benchmark (Client Dataset).

3. Cross-Dataset Testing

After individual training, each model was tested on all team datasets to evaluate cross-dataset generalization. This simulates a real-world forensic scenario where a trained detector must analyze unseen deepfake content from different sources.

3.1 Comparison Table

The table below summarizes all training and cross-dataset testing results. Each row represents one algorithm, showing its training accuracy (ACC) and ROC-AUC performance on each dataset.

Algorithm	**Training Dataset**	**Train Acc**	**Test Acc**	**FF++**	**Celeb-DF V2**	**UADFV**	**CIFAKE**	**Client Dataset**
EfficientNet-B4	FF++ C23	ACC 93% AUC: 0.9511	ACC 86.78% AUC: 0.9519	ACC 69.62% AUC: 0.7656	ACC 53.57% AUC: 0.5673	ACC 58.03% AUC: 0.6398	ACC 52.12% AUC: 0.5228	ACC 47.39% AUC: 0.4681
MesoNet	UADFV	ACC 95.16% AUC 0.9841	ACC 95.5% AUC 0.9655	ACC 50.15% AUC 0.5024	ACC 51.02% AUC 0.5417	ACC 95.5% AUC 0.9655	ACC 38.29% AUC 0.3450	ACC 48.53% AUC 0.4695
XceptionNet	CelebDF_V2(image Dataset)	ACC. 75.92% AUC 0.8411	ACC. 76.26% AUC 0.8555	ACC. 50.72% AUC 0.5183	ACC. 76.26% AUC 0.8555	ACC. 58.77% AUC 0.5981	ACC. 50.22% AUC 0.5587	ACC. 49.41% AUC 0.4748
ResNet-50	CIFAKE	ACC. 97.80% AUC 0.9975	ACC. 97.80% AUC 0.9975	ACC. 50.00% AUC 0.4992	ACC. 49.90% AUC 0.4070	ACC. 50.21% AUC 0.3300	ACC. 97.80% AUC 0.9975	ACC. 50.00% AUC 0.4734
ViT	OpenForensics	ACC. 80.24% AUC 0.8944	ACC 88.26% AUC: 0.9234	ACC. 49.88% AUC 0.4943	ACC. 51.86% AUC 0.5571	ACC. 49.88% AUC 0.4943	ACC. 44.34% AUC 0.4173	ACC. 54.55% AUC 0.5638

Note: All accuracy(ACC) values are rounded to 2 decimal places. AUC-ROC values are shown where available.

EfficientNet-B4

Figure 1 below illustrates EfficientNet-B4's accuracy and AUC-ROC across all tested datasets, showing a consistent decline as the model is evaluated on increasingly different domains from its training data.

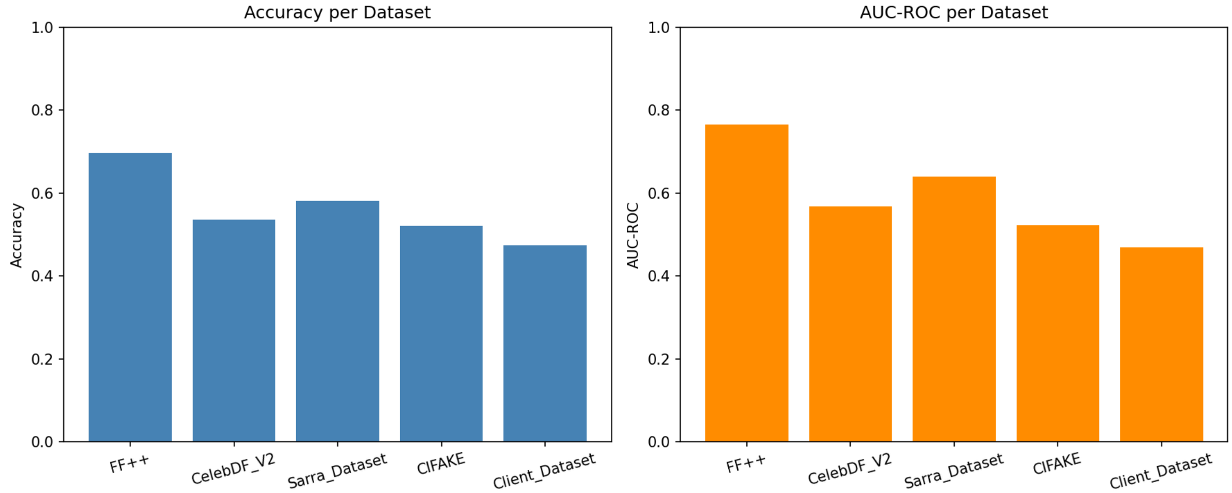

Figure 1: EfficientNet-B4 Cross-Dataset Performance - Accuracy and AUC-ROC Comparison

ResNet

Figure 3 below illustrates the ResNet performance metrics, demonstrating high precision in detecting Stable Diffusion artifacts within CIFAKE. However, a significant generalization gap is evident, as accuracy drops to approximately 50% across all external deepfake datasets.

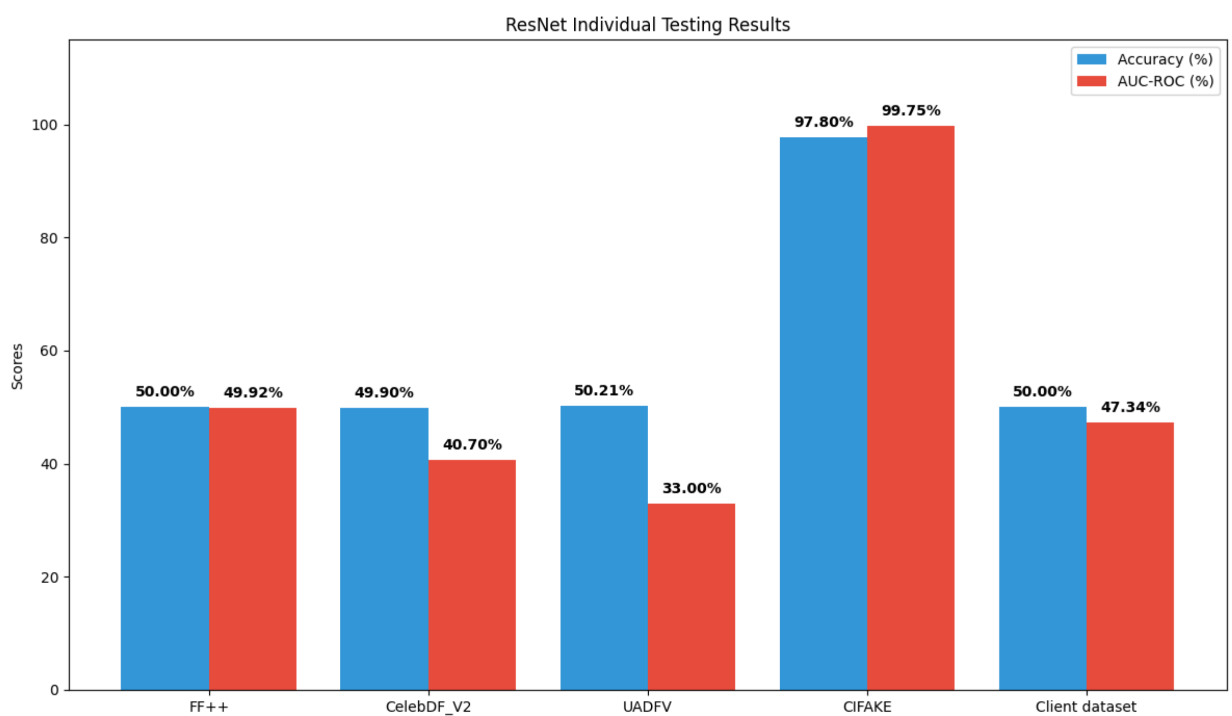

Figure 3: ResNet Cross-Dataset Performance

TiV

Figure 5 presents the accuracy and AUC-ROC of the ViT model across multiple unseen data. The model performs best on OpenForensics testing data and drops on other data. However, results remain relatively stable, demonstrating reasonable generalization ability.

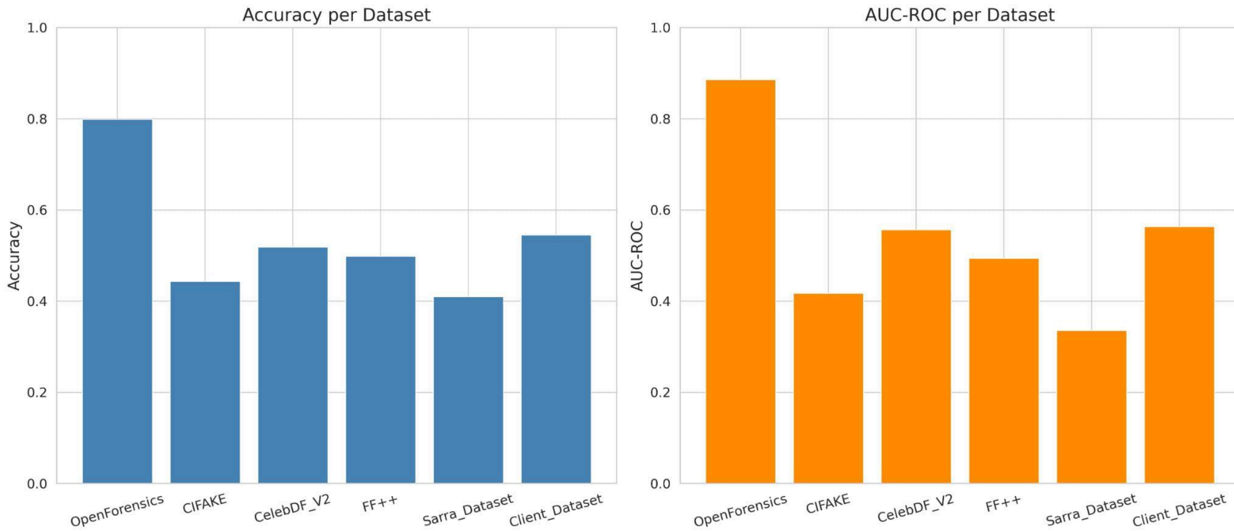

Figure 5: Cross-Dataset Performance of ViT for deepfake detection

Mesonet

Figure 6 presents the accuracy and AUC-ROC of the Mesonet model across multiple unseen data. The model performs extremely well on UADFV because it learned dataset-specific patterns on training, However, its performance drops significantly on other datasets, with results close to random classification.

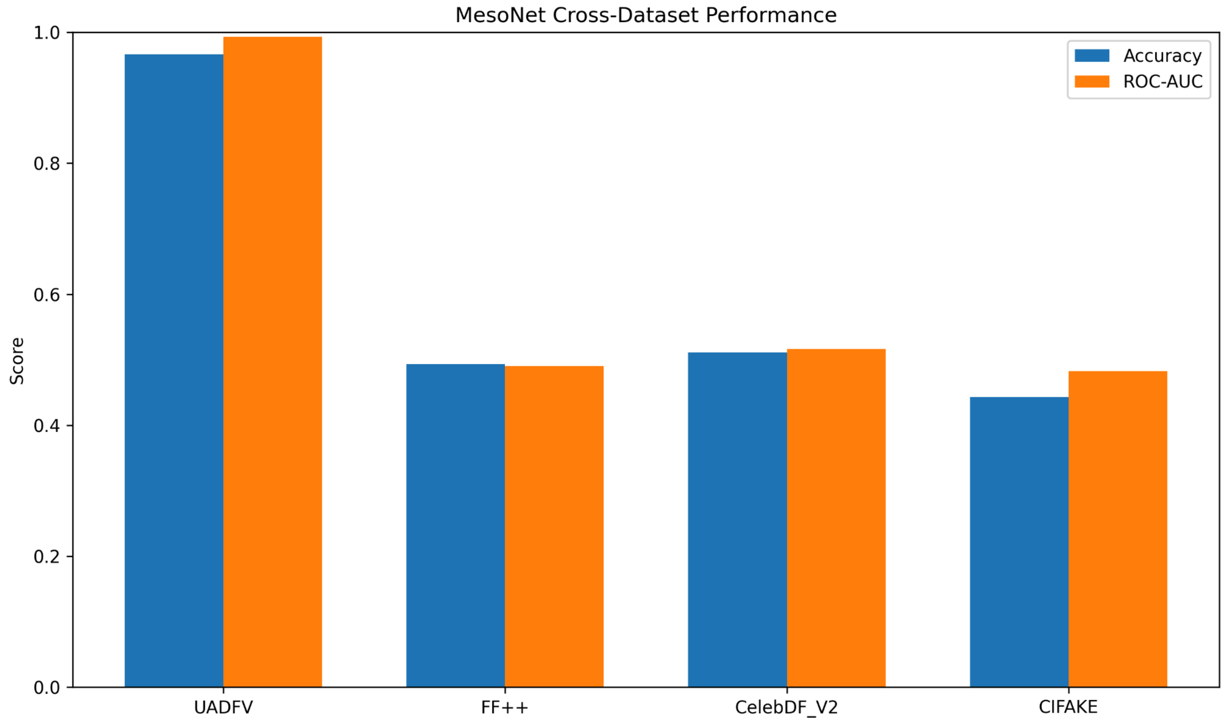

Figure 6 : Cross-Dataset Performance of Mesonet for deepfake detection

XceptionNet

As observed in the results shown in the table below, the model performed best on its own test set with 76.26% accuracy and a strong AUC-ROC of 0.8555. The model's performance dropped significantly on the other datasets, having an accuracy between 49% and 59%. This shows that the model has limitations such as limited generalization when the deepfake generation method, image quality or source is different from its training dataset (CelebDF_V2(image Dataset)). This is a common challenge in deepfake detection.

Dataset	**Accuracy**	**AUC-ROC**	**F1-Score**	**Log Loss**
CelebDF_V2(image Dataset) (own test set)	0.7626	0.8555	0.7277	0.4741
FaceForensics++ C23	0.5072	0.5183	0.6453	2.1960
UADFV	0.5877	0.5981	0.6499	0.9444
CIFAKE	0.5022	0.5587	0.6661	3.4846
140k Real and Fake Faces (client dataset)	0.4941	0.4748	0.6583	5.1242

4. Analysis and Discussion

4.1 Cross-Dataset Performance Analysis: Why Each Algorithm Succeeds and Fails

4.1.1 EfficientNet-B4 - Trained on FaceForensics++ (FF++)

EfficientNet-B4 was trained on FF++ C23 and learned to detect compression-specific artifacts and face swap patterns unique to this dataset. Its performance across all datasets is as follows:

- **FF++ (69.62% Accuracy, 76.56% AUC)** - Best performance as expected. The model was trained on this dataset and is familiar with its H.264 C23 compression artifacts, face swap boundaries, and manipulation patterns from four methods (DeepFakes, Face2Face, FaceSwap, NeuralTextures). The score is lower than training accuracy because cross-dataset testing used all 30,000 images including harder unseen examples, compared to the small controlled 15% test split used during training.
- **CelebDF V2 (53.57% Accuracy, 56.73% AUC)** - Poor performance. CelebDF uses significantly higher quality face swaps with advanced blending, color correction, and post-processing that the model never encountered during training. The visual characteristics of celebrity faces with professional lighting and makeup are also very different from FF++ video frames, leaving the model with no reliable signal to detect fakes.
- **UADFV (58.03% Accuracy, 63.98% AUC)** - Slightly above random. UADFV uses older and simpler face swap techniques that leave more obvious visual artifacts, somewhat similar in nature to FF++ manipulation artifacts. The model accidentally picks up some of these familiar patterns, resulting in marginally better performance compared to other unseen datasets.
- **CIFAKE (52.12% Accuracy, 52.28% AUC)** - Near random guessing. CIFAKE contains images generated by Stable Diffusion, not face swaps. The model was never trained on AI-generated image artifacts and has no knowledge of diffusion model fingerprints, making it completely ineffective on this dataset.
- **Client Dataset - 140k Real vs Fake (47.39% Accuracy, 46.81% AUC)** - Worst performance, below random guessing. The client dataset contains StyleGAN generated faces - hyper-realistic AI images of people who do not exist. The model not only fails to detect them but actively predicts the wrong label more often than not, confirming that StyleGAN artifacts are completely outside the face swap domain EfficientNet-B4 was trained on.

4.1.2 MesoNet - Trained on UADFV

the model overfits to the training dataset and fails to generalize to unseen data. These findings highlight the limitations of using a single dataset for training in real-world deepfake detection scenarios.
Its performance across all datasets is as follows:

- **UADFV (95.5% Accuracy, 96.55% AUC)** - MesoNet performs very well on UADFV because it was trained on this dataset. The deepfake manipulations in UADFV are relatively simple and contain visible artifacts such as warping and blending inconsistencies. The dataset also has low variability in terms of lighting, compression, and facial features. The high performance is due to dataset familiarity and overfitting, not strong generalization.

- **FF++ (50.15% Accuracy, 50.24% AUC)** - Performance drops significantly on FF++ due to heavy compression (C23) and the presence of multiple manipulation techniques. Compression removes many of the visual artifacts that MesoNet relies on, making detection more difficult.
- **CelebDF V2 (51.02% Accuracy, 54.17% AUC)** -CelebDF contains high-quality and realistic deepfakes with minimal visible artifacts. The faces are better aligned and blended, making manipulations harder to detect. MesoNet struggles because it relies on obvious artifacts, which are largely absent in this dataset.
- **CIFAKE (38.29% Accuracy, 34.50% AUC)** -Worst performance, CIFAKE is fundamentally different from other datasets, as it contains fully AI-generated images rather than face-swapped videos. The types of features learned from UADFV are not relevant to this dataset.
- **Client Dataset - 140k Real vs Fake (48.53% Accuracy, 46.95% AUC)** - The client dataset represents real-world conditions with unknown characteristics, including variations in lighting, resolution, and manipulation techniques. The model has not seen similar data during training.

4.1.3 XceptionNet - Trained on CelebDF_V2(image Dataset)

XceptionNet was trained on CelebDF_V2(image Dataset) based on the original Celeb-DF (V2) video dataset. It learned to detect small changes such as face-swap artifacts, blending inconsistencies, and high-resolution manipulation patterns specific to this dataset. Its performance across all datasets is as follows:

- **CelebDF_V2 (own test set) (76.26% Accuracy, 0.8555 AUC)** - Best performance as expected. The model was trained on this exact dataset and became familiar with its realistic face swaps and minimal visible artifacts. This allowed it to detect deepfakes effectively within the same distribution.
- **FF++ (50.72% Accuracy, 0.5183 AUC)** - Close to random guessing. FF++ contains heavily compressed videos (c23) with multiple manipulation methods and visible compression artifacts. These characteristics are very different from the high-quality and clean images in CelebDF_V2 so the model's learning did not perform well on this dataset.
- **UADFV (58.77% Accuracy, 0.5981 AUC)** - Slightly above random. UADFV uses older and simpler deepfake techniques with more obvious warping and blending errors. These basic artifacts slightly overlap with what the model saw during training which gives a small advantage of detecting deepfakes better in this dataset than other more complex datasets.
- **CIFAKE (50.22% Accuracy, 0.5587 AUC)** - Close to random guessing. CIFAKE consists of fully AI-generated images created with Stable Diffusion and not face-swapped images. Since the model was trained only on real face-swap manipulations, it doesn't know anything about synthetic textures and diffusion artifacts present in CIFAKE.
- **Client Dataset - 140k Real vs Fake (49.41% Accuracy, 0.4748 AUC)** - Worst performance and below random guessing. The client dataset contains a wide variety of unknown manipulation techniques and real-world variations. These conditions are different from the high-quality celebrity face swaps in CelebDF_V2 which leaves the model with no hints on whether an image is a deepfake or not.

4.1.4 ResNet-18 - Trained on CIFAKE

ResNet-18 achieved a near-perfect 97.80% accuracy and 0.9975 AUC on its native CIFAKE dataset, demonstrating its high proficiency in identifying the specific high-frequency "fingerprints" and mathematical noise patterns unique to Stable Diffusion generation. However, the model suffered a total generalization failure across all other benchmarks, averaging approximately 50% accuracy, which is equivalent to random guessing.

This stark drop-off occurs because the model was trained on purely synthetic textures rather than the facial blending or boundary inconsistencies found in video-based datasets like FF++, Celeb-DF V2, and UADFV. Furthermore, its failure on the Client Dataset (50.00% accuracy) confirms that diffusion-based detection features do not translate to StyleGAN artifacts, proving that the model is a highly effective specialist for its training domain but lacks the robustness to detect diverse generative or face-swap manipulation methods.

4.1.5 ViT - Trained on OpenForensics

ViT achieves its best performance on OpenForensics testing data, with an accuracy of 88.26 % and 0.9234 AUC-ROC (even though this test data was not seen during training). This indicates in-domain generalization [7], as the training and testing sets share similar characteristics such as image quality and compression settings [1]. This performance is expected as prior works show that models typically perform well when train and test data come from the same distribution [1].

Performance decreases on unseen datasets, starting with the Global 140k Client Dataset (54.55%) and Celeb-DF V2 (51.86%), where the model still maintains moderate accuracy. That likely happens due to some overlap in facial features and manipulation patterns. It can also occur due the differences in data collection and synthesis pipelines that might introduce distribution shift [13].

Furthermore, the model drops on UADFV (Sarra_dataset), which achieves 50.60%, FF++ at 49.88%, and especially CIFAKE at 44.34%, where performance falls below 50%. These results show the impact of domain shift (e.i., concept drift), including variations in generation methods, image quality, and post-processing. Such differences might prevent the model from effectively generalizing out-of-distribution data. This limitation is a major challenge and widely recognized, as deepfake detection models often rely on dataset-specific artifacts rather than learning universal forgery [12]. The corresponding AUC-ROC values follow a similar trend, decreasing from 0.9234 (OpenForensics) to 0.4173 (CIFAKE), further confirming reduced discriminative ability under cross-dataset conditions.

Overall, the results demonstrate that while the model generalizes well within similar domains and patterns, its performance degrades under significant distribution shifts, emphasizing that cross-dataset generalization remains a key challenge in deepfake detection.

4.2 Which Algorithm Performed Best Overall and Why

Based on cross-dataset results, no single algorithm can be declared the definitive best. Each model performed well only on its own training dataset and struggled on all others. However this is not a limitation of the algorithms themselves - it reflects a well-documented challenge in deepfake detection research known as domain shift.

Looking at average cross-dataset accuracy excluding each model's own dataset, EfficientNet-B4 (52.78%) and XceptionNet (52.28%) show marginally better generalization compared to MesoNet (46.99%), ResNet-18 (50.03%) and ViT (~47%). The differences are small and expected - each model was trained on a single dataset with access to only one type of deepfake pattern. Tolosana et al. [11] confirm that cross-dataset generalization remains one of the most persistent challenges in deepfake detection, even for state-of-the-art systems.

Each algorithm successfully learned and detected manipulation patterns in its training dataset with strong accuracy, confirming correct implementation and sound methodology. As Dolhansky et al. [10] showed in the DeepFake Detection Challenge, even models trained on large diverse datasets experience significant performance drops on unseen deepfake types - our results are consistent with this.

The cross-dataset drop highlights exactly why real cybercrime investigations require diverse training data and multi-model approaches. Each of our five algorithms contributes a specialized capability, and together they provide a broader empirical understanding of how different deepfake generation methods behave under cross-domain evaluation.

5. Best Algorithm Recommendation

Based on the cross-dataset evaluation conducted in this project, recommending a single best algorithm for all forensic deepfake detection scenarios is not straightforward. Each model demonstrated strong performance on its own training domain but generalized poorly to unseen datasets, reflecting the inherent limitations of single-dataset training rather than any shortcoming in the algorithms themselves.

To determine which algorithm generalizes best, we calculated the average accuracy on unseen datasets only, excluding each model's own training dataset from the comparison:

Algorithm	Average Accuracy on Unseen Datasets
EfficientNet-B4	52.78%
XceptionNet	52.28%
ViT	50.10%
ResNet-18	50.03%
MesoNet	46.99%

It should be noted that dataset sizes vary significantly across evaluations, ranging from 946 images in UADFV to 30,000 in FF++, and these averages should therefore be interpreted as indicative rather than definitive rankings.

With that context in mind, EfficientNet-B4 and XceptionNet show the most consistent generalization across unseen datasets. For a forensic investigation involving video-based face swap evidence, EfficientNet-B4 would be the recommended starting point given its training on FaceForensics++, the most diverse and widely used face manipulation benchmark. For high quality GAN-based deepfakes, XceptionNet would be more appropriate given its training on CelebDF V2. For cases involving AI-generated synthetic images rather than face swaps, ResNet-18 or ViT would be better suited.

The most reliable forensic recommendation however is a multi-model approach - applying multiple specialized detectors to the same evidence and cross-verifying outputs - as no single algorithm evaluated here consistently exceeds 55% accuracy on truly unseen deepfake types.

6. Forensic Investigation Scenario

6.1 The Case: Client Dataset

In this project, the client provided a dataset containing images labeled as real and fake, representing a potential cybercrime scenario involving manipulated digital media. The objective was to determine whether the provided media had been altered using deepfake techniques.
This scenario simulates a real-world investigation where digital evidence must be analyzed to verify authenticity. The dataset represents unknown conditions, making it suitable for evaluating the robustness of deepfake detection models.

6.2 Algorithm Recommendation

Based on the experimental results, while ViT is recommended as a supporting model, as it achieved the best performance on the client dataset, EfficientNet-B4 is recommended as the primary detection algorithm. Although it does not achieve the highest accuracy on any individual dataset, it demonstrates the most consistent performance across multiple datasets (FF++, Celeb-DF, UADFV, CIFAKE) without catastrophic degradation. This indicates that EfficientNet-B4 is less reliant on dataset-specific artifacts (e.g., compression signatures or GAN fingerprints) and instead captures more generalizable features relevant to deepfake detection. However, best practices and recent research suggest employing an ensemble of different models (ensemble-based approach) instead of trusting one model only [13]. Additionally, for legal proceedings, AUC is more meaningful than accuracy because it captures the model's ability to rank real vs. fake samples across decision thresholds. Accuracy at a fixed threshold can be misleading [18].

6.3 How We Process the Evidence

The investigation follows a structured digital forensics workflow:

Data Acquisition: The client dataset is securely collected and preserved to maintain integrity.

Preprocessing: Images are resized and converted into a format compatible with the trained models.

Model Inference: The trained deepfake detection model is applied to classify each image as real or fake.

Feature Analysis: The model identifies inconsistencies such as unnatural textures or blending artifacts.

Result Interpretation: Outputs are analyzed using: Accuracy , F1-score ,ROC-AUC , Confusion matrix

Reporting: Findings are documented and prepared for legal or investigative use.

6.4 Presenting in Court: Expert Witness Framework

In a legal context, deepfake detection models cannot independently determine authenticity or guilt. Instead, they serve as **supporting forensic tools** that assist human experts.

As expert witnesses, our role is to present the findings in a clear, transparent, and scientifically valid manner. This includes:

- **Explaining the methodology**
  Describing how the model was trained, tested, and applied to the client dataset in a way that is understandable to non-technical audiences.
- **Presenting quantitative evidence**
  Reporting performance metrics such as accuracy, F1-score, and ROC-AUC to demonstrate the reliability of the model.
- **Providing visual evidence**
  Including confusion matrices and prediction examples to illustrate model behavior.
- **Highlighting limitations**
  Clearly stating that the model achieved approximately 54.38% accuracy on the client dataset, indicating limited reliability.
- **Ensuring transparency**
  Acknowledging uncertainties, potential biases, and the impact of dataset differences (domain shift).

Due to the relatively low performance on the client dataset, the model's output should not be considered definitive proof, but rather supporting evidence that must be corroborated with additional forensic analysis.

6.5 Applicable Canadian Law

In Canada, digital evidence must meet standards of **reliability, authenticity, and admissibility**.

Relevant legal principles include:

Canada Evidence Act : governs admissibility of digital evidence [3]

R. v. Mohan (1994) : defines criteria for expert witness testimony: Relevance, Necessity, Absence of bias, Proper qualifications [4]

To be admissible, deepfake detection methods must be:

- Scientifically valid
- Properly documented
- Reproducible
- Interpreted by qualified experts

Given the model's limited performance on the client dataset, its results must be used with caution and supported by additional forensic evidence to meet legal standards.

7. Conclusion

This project evaluated five deepfake detection algorithms. Each algorithm was trained on only one dataset and then tested on all the other datasets including datasets that other models trained on respectively and a dataset that no model trained on. The results showed a clear pattern where every model performed well on the dataset it was trained on but performed poorly on completely new datasets it had not encountered during training. When a model is tested on its own test set, it is seeing images that are very similar to what it learned during training. On these unfamiliar datasets, the model's performance would often be close to 50% accuracy which is almost the same as random guessing.

This outcome is normal and probably as expected since each model only learned the specific patterns and artifacts present in its single training dataset. It never saw the different manipulation techniques, images qualities or compression levels found in the other datasets. When tested on other datasets with deepfakes created differently or with varying image qualities, the model's performance would drop significantly which indicates that the model is not able to tell which images are real or fake in a reliable manner. This is known as the cross-dataset generalization problem and it is a well-documented limitation in deepfake detection research [11], [15].

In real cybercrime cases, investigators would often receive deepfake images that come from unknown sources and the methods used to create these deepfakes are also unknown. These are the types of datasets that our models struggled with, datasets where deepfakes are unfamiliar or come from unknown sources.

Thus, there is no single "best" algorithm among the five we tested because of this limitation (cross-dataset generalization). Each model works well only when the images are similar to the data it was trained on. In practice, investigators would either need models that were trained on many different deepfake datasets at once or they would need to combine many specialized models and compare their outputs [10].
This project demonstrates that deepfake detection remains an evolving field. As criminals develop more advanced and realistic deepfakes especially with the continuous evolution of AI. Detection methods must also continue to improve by training models on larger and more diverse datasets, using many models together to process deepfakes and always having human experts to review the results.

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Datasets

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